AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM351
Data Management and Analysis

Final Examination
Fall Semester 2024/2025

Date: 11/Jan./2025

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Number of Exam Pages: 4 (including this cover sheet)	Time Allowed:3 Hours

Instructions:

- Total Marks: 100
- This exam consists of 3 parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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Part One: [10 marks]

Please select **the best and most appropriate answer** and write it on the answer booklet.

1. Object relational databases supports the following data base concepts.
a. Encapsulation b. Inheritance c.Both d. none of the above
2. In single master transaction data may be shared across _____.
a. Multiple Nodes b. Single Nodes c. non of the above d. all of the above
3. Classification tasks like identifying email spam and classifying credit applications, which use la-belled training data to try to classify unseen instances, are known as _____ tasks
a. regression b. unsupervised learning c. supervised learning d. clustering
4. A relation is in Third Normal Form if all the non-key columns are dependent only on _____
a. primary key b. foreign key c. Both d. none of the above
5. Which of the following descriptive statistical techniques requires no training data ?
a. classification b. clustering c. KNN d. all of the above
6. Replication is mostly concerned with data resilience whereas _____ is concerned with capacity and performance.
a. sharding b. centralisation c. the user interface d. none of the above
7. _____ is concerned with segmenting a diverse group of data into a number of similar sub-groups.
a. clustering b. correlation c. combination d. regression
8. _____ is part of data integration and is concerned with ensuring domain consistency.
a. Format integration b. Semantic Integra-tion c. Data cleansing d. Data scrubbing
9. A relation is in First Normal Form (1NF) if each attribute contains only _____ values, that is, it has no repeating groups of values.
a. atomic b. numerical c. string d. none of the above
10. Which of the following descriptive statistical techniques requires no training data ?
a. list b. dict c. dataframe d. none of the above

Part Two: [90 marks]

Question One: [30 marks]

A database stores information about supplier, item, item_supplier, and item_category. The informatio about the supplier is kept in the supplier relation. Item relation contains the details about the items. Item_supplier relation contains the details of the items supplied by a supplier. Item_category relation contains the details about the categories of items. The relations are given below:

Supplier (**supplier_code**, firstName, lastName, supplier_city, supplier_phone)
Item(**item_code**, item_name, *item_category_code*, item_description, quantity_in_stock)
Item_supplier(***item_code, supplier_code***, date_of_supply, quantity_supplied, price)
Item_category(**item_category_code**, item_category_description)

Use the above relations and write SQL statements for the following questions: -

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a. Write an SQL query to find all the items issued by the supplier having first name 'Talal'. Return item code, date of supply, quantity supplied and price. [use SUBQUERY] [5 marks]

b. Write an SQL query to find the highest quantity supplied by each supplier. Filter the result by the highest quantity supplied above 100. Return supplier code and maximum quantity supplied. [5 marks]

c. Write an SQL query to find the suppliers who stay in the same city as the supplier with the last name 'Al Balushi'. Return first name and last name. [5 marks]

d. Write an SQL query to find those items whose name does not contain the letter 'M'. Sort the result-set in ascending order by item code. Return item code and item name. [5 marks]

e. Write an SQL query to find all the items from the category code 'IC009' and whose quantity in stock is more than 150. [5 marks]

f. Write an SQL query to increase the price of item with code 'IT1' supplied by supplier with code 'SUP01' by 2%. [5 marks]

Question Two: [20 marks]

Use the sample below from the **student** document database collection to answer the following questions: (assume the database name is: "**db**" and the collection name is: "**student**").

```
{
   "student_id": "1234",
   "fname": "Muna",
   "lname": "Al Busaidi",
   "address": {
      "building": "107",
      "street": "Park Ave",
      "zipcode": "10462"
   },
   "student_level":"Diploma",
   "department":"IT",
   "grades": [
{"courseID":"T351","tot_marks": 91,"grade": "A"},
{"courseID":"T101","tot_marks": 83,"grade": "B"},
{"courseID":"T359","tot_marks": 82,"grade": "B"},
{"courseID":"T160","tot_marks": 69,"grade": "C"}
   ]
}
```

a. Write a MongoDB query to display all the documents in the collection student. [2 marks]
b. Write a MongoDB query to display all the students who are in IT department. [3 Marks]
c. Write a MongoDB query to display the first five students in Diploma level. [4 marks]
d. Write a MongoDB query to find the students who achieved an 'A' grade. [5 marks]

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e. Write a MongoDB query to find the students who achieved total marks more than 80 an than 95. **[6 marks]**

Question Three: [15 marks]

An experiment involves two groups of participants, Group A and Group B and we have recorded the grade each group member for two courses, M251 and TM351 as follows:

Group A

StudentID	M251	TM351
A1	75	80

Group B

StudentID	M251	TM351
B1	60	70

Assume that a new student, S1 whose grade of M251 is 62 and grade of TM351 is 75 who needs to be plac into either Group A or Group B. Calculate the similarity ratio of S1 with A1 and B1.

Question Four [25 marks]

a. What is Indexing? What is the use of Indexing in SQL? [5 marks]

b. Represent the following documents by their term vectors. [5 marks]

Doc1: The bird's nest is best.

Doc2: The bird's nest is not good.

c. Write a short note on cosine similarity. How to calculate the cosine angle between two documents? [5 marks]

d. What is cluster cohesion and cluster separation? What is its use? [5 marks]

e. Can normalization avoid 100% redundancy? Please justify your answer. [5 marks]

End of Questions

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